Jane Yang

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RESEARCH INTERESTS

Egocentric vision, visual representation learning, and perception and interaction in naturalistic environments.

EDUCATION

Ph.D. in Experimental Psychology, UC San Diego. Advisor: Bria Long 2024–present

B.S. in Cognitive Science, UC San Diego. Machine Learning & Neural Computation; minor in Computer Science 2018–2022

AWARDS & FELLOWSHIPS

National Eye Institute Early Career Scientist Travel Grant, Vision Science Society 2026
Anderson Travel & Research Award, UC San Diego 2025
Norman Henry Anderson Fellowship, UC San Diego 2024
Triton Research and Experiential Learning Scholars Award, UC San Diego 2022
Distinction in Cognitive Science, UC San Diego 2022
Provost's Honors, UC San Diego 2019–2022
HackSC 1st Place in Entrepreneurship, USC 2019

TECHNICAL SKILLS

Programming	Python, MATLAB, R, C++, C, JavaScript, TypeScript, Java, Clojure
Machine learning	PyTorch, TensorFlow, HuggingFace
Data processing	Multi-sensor fusion, video processing, SLURM cluster computing, Docker, conda environment management
Data collection	Head-mounted camera and eye tracking, depth cameras, IMU, multi-device synchronization
Development	Git, CI/CD, MongoDB, Node.js, Unity, Blender, AutoCAD, SolidWorks
Other software	LaTeX, Datavyu, Audacity, ELAN, Illustrator
Languages	English, Mandarin, Hokkien

PUBLICATIONS

Manuscripts

Yang, J., Sepuri, T., Tan, A.W.M., Aw, K.L., Frank, M.C., & Long, B. (2026). Characterizing the visual representation of objects from the child's view. *Under review*. [link]

Tan, A.W.M.*, **Yang, J.***, Sepuri, T., Aw, K.L., Sparks, R.Z., Yin, Z., Marchman, V.A., Frank, M.C., & Long, B. (2025). Assessing the alignment between infants' visual and linguistic experience using multimodal language models. *Preprint*. [link]

Yu, Z., Aubret, A., Raabe, M.C., **Yang, J.**, Yu, C., & Triesch, J. (2025). Toddlers' active gaze behavior supports self-supervised object learning. *Developmental Science*. [link]

Peer-reviewed Conference Papers

Yang, J., Sepuri, T., Tan, A.W.M., Aw, K.L., Frank, M.C., & Long, B. (2026). Variability in young children's everyday visual experiences of object categories. *Cognitive Computational Neuroscience*. [link]

Yang, J., Sepuri, T., Tan, A.W.M., Aw, K.L., Frank, M.C., & Long, B. (2026). Characterizing the inputs to infants' object category representations. *Vision Science Society*. [link]

Yang, J., Sepuri, T., Tan, A., Frank, M.C., & Long, B. (2025). Quantifying infants' everyday experiences with objects in a large corpus of egocentric videos. *Cognitive Computational Neuroscience*. [link]

Yang, J., Zhang, Y., & Yu, C. (2024). Learning semantic knowledge based on infant real-time attention

and parent in-situ speech. *Cognitive Science Society*. [link]

Yang, J., Smith, L., Crandall, D., & Yu, C. (2023). Using manual actions to create visual saliency: an outside-in solution to sustained attention and joint attention. *Cognitive Science Society*. [link]

Wang, H., **Yang, J.**, Tamari, R., & Fan, J. (2022). Communicating understanding of physical dynamics in natural language. *Cognitive Science Society*. [link]

SELECTED PRESENTATIONS

Invited talk, Early Career Speaker Series, Cognitive & Brain Sciences, University of Nevada, Reno 2026

Variability in young children's everyday visual experiences of object categories 2026
Spotlight poster: Cognitive Computational Neuroscience.

Characterizing the inputs to infants' object category representations 2026
Talks: Beyond Curated Datasets (CCN satellite); Vision Science Society; Southern California Meeting for Investigations in Developmental Science; Egocentric Video Workshop, Stanford.

Quantifying infants' everyday experiences with objects in a large corpus of egocentric videos 2025
Talks: BabyView Workshop (Understanding Infant Experience Through Egocentric Data); BabyView and 1kD joint lightning talks. Posters: Cognitive Computational Neuroscience; Southern California Meeting for Investigations in Developmental Science.

Learning semantic knowledge based on infant real-time attention and parent in-situ speech 2024
Talks: Cognitive Science Society (46th Annual Meeting); International Congress of Infant Studies.

Using manual actions to create visual saliency 2023
Talk: Cognitive Science Society (45th Annual Meeting). Posters: COSMOS; Workshop on Natural Environments, Tasks and Intelligence.

Communicating understanding of physical dynamics in natural language 2022
Talk: UCSD 35th Annual Undergraduate Research Conference. Poster: Cognitive Science Society (44th Annual Meeting).

RESEARCH EXPERIENCE

Graduate Student, UC San Diego Visual Learning Lab. PI: Bria Long 2024–present
Overarching question: how the visual input children actually receive shapes the object representations they build, and how that input compares to what vision models are trained on.

- **What objects children see, and how that differs from curated datasets.** Object frequency and representational geometry across 1,482 hours of egocentric video (BabyView, 35 families) compared against adult egocentric experience (VEDB) and THINGS. *Methods:* YOLOE detection with CLIP-based filtering, now moving to a locally run Qwen vision-language model validated against a human-annotated gold set; CLIP and DINOv3 embeddings, representational similarity analysis. *Output:* first-author manuscript under review; public reproducibility capsule.
- **Why some categories are visually harder than others.** Global and local dispersion measures over 85 categories separate viewpoint-driven from instance-driven variability; rankings partially replicate across backbones and are only weakly coupled to frequency. *Output:* CCN 2026 spotlight; reproducible analysis package and layer-wise embedding explorer.
- **Whether children and models converge on 3D shape.** MOCHI Kids, a developmental (ages 4–6) adaptation of the MOCHI shape-oddity benchmark, built and deployed for direct human-model comparison on multiview object consistency. *Methods:* TypeScript and jsPsych client, Express and MongoDB backend, end-to-end tests in CI. *Status:* piloting.
- **How language and vision align in everyday input.** Alignment between child-directed speech and co-occurring visual scenes in naturalistic video, using multimodal models as the measure. *Output:* co-first-author preprint.
- **What changes when children act on objects.** Identifying viewed and manipulated objects by combining monocular depth estimation, IMU-derived head direction, and hand tracking; building a synchro-

nized capture rig (Pupil Labs Neon eye trackers, two depth cameras, room-view camera) for caregiver-child interaction studies. *Status:* in progress.

Lab Technician, UT Austin Developing Intelligence Lab. PI: Chen Yu 2022–2024

- **Multimodal measurement of infant-caregiver interaction.** Ran home-based sessions with 12–36 month infants using head-mounted eye trackers, third-person cameras, IMUs, and lapel microphones. *Methods:* Python preprocessing pipelines across devices; Built a MATLAB codebase to analyze and visualize multi-stream data; YOLOv8 fine-tuning for infant and parent views; MediaPipe and OpenPose for hand and face detection; WhisperX for transcription and diarization; skeletal tracking for dyadic play; 3D environments in Unity, Blender, and Matterport.

Research Assistant, UC San Diego Cognitive Tools Lab. PI: Judith Fan 2021–2022

- **How people communicate abstract physics knowledge.** Honors thesis. *Methods:* web-based experiments in JavaScript and Node.js; NLP and manual annotation of 480+ participant responses.

Research Assistant, UC San Diego Language and Comprehension Lab. PI: Eva Wittenberg 2020–2021

- **How verbal reduplication shapes event conceptualization in Mandarin.** *Methods:* web-based psycholinguistic experiments; development of a pressure-sensing pen for physiological response recording.

SOFTWARE & OPEN SCIENCE

MOCHI Kids. Kid-friendly web implementation of the MOCHI 3D shape-oddity benchmark for human-model comparison. TypeScript, jsPsych, Express, MongoDB, end-to-end tests in CI.

BabyView Objects reproducibility capsule. Analysis code, anonymized data, and a pinned Docker environment reproducing the manuscript results. MIT and CC0 licensed.

CCN 2026 object variability package. Reproducible dispersion metrics and figures over 7,018 validated crops with paired CLIP and DINOv3 embeddings.

BabyView variability explorer. Layer-wise reanalysis of CLIP dispersion metrics with an interactive embedding explorer across transformer blocks.

TEACHING

UC San Diego, Department of Psychology

PSYC 201A Statistical Computing and Inference from Data I	2026
PSYC 201B Quantitative Methods in Psychology II	2026
PSYC 201A Statistical Computing and Inference from Data I	2025
PSYC 168 Psychological Disorders of Childhood	2025
PSYC 181 Drugs and Behaviors	2025
PSYC 102 Sensory Neuroscience	2024

UT Austin, Department of Psychology

PSY 371M Introduction to Machine Learning	2023

UC San Diego, Department of Cognitive Science

COGS 189 Brain Computer Interfaces	2022
COGS 101C Language	2021

MENTORSHIP

UC San Diego: Dora Deng, Mira Mateo, Juan Pablo Bello, Rochelle Boegeman, Emma Nishikawa 2024–
UT Austin: Elton Martinez, Anagha Kenikar, Jacob Rivera, Ruchi Shah 2023–2024

SERVICE & LEADERSHIP

Co-organizer, *Beyond Curated Datasets* CCN satellite workshop, New York, NY 2026
Colloquium committee representative, Department of Psychology, UC San Diego 2025–2026